Sony patented VR controller
A new patent has been filed by Sony for new VR motion controllers that would probably be making their way into the PlayStation VR ecosystem. **http://dgit.in/SonyVRCntrl**

New Spectre fix
Recent Spectre fixes were apparently buggy so Microsoft released new Windows patch to disable those fixes. **http://dgit.in/MsftSpctr**

phase-change cooler is a unit which usually sits underneath the PC, with a tube leading to the processor. Inside the unit is a compressor of the same type as in an air conditioner. As in a fridge, the compressor compresses the gas into a liquid which is then pumped to the processor, at which point it passes through a condenser then an expansion device which evaporates it again, thus dispersing the heat. This type of a cooling system can be expensive, and is hence not found in mainstream implementations.

Slightly more mainstream among enthusiasts is the use of liquid nitrogen to cool overclocked machines. Due to its extremely low boiling point of -196°C, it is highly effective coolant for short overlocking sessions. In a typical setup, a copper or aluminium pipe is installed over the processor or graphics card that is to be overclocked. After proper insulation, the liquid nitrogen is poured into the pipe. While there are devices to hold the poured liquid nitrogen, once it evaporates it has to be refilled. Additionally, although it is not flammable, it condenses oxygen quite rapidly from the atmosphere, which is highly flammable itself.

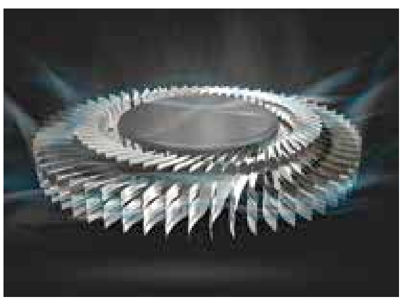
Liquid nitrogen being used during overclocking

LAPTOPS

When it comes to designing cooling solutions for laptops, both the portability and the computing power have to be taken into account. It wouldn't be too wrong to say that laptops have the constraints of both smartphones and PCs when it comes to thermal management. In fact, that is one of the main reasons why laptops, except for a few high end ones, still offer mostly rudimentary thermal management capabilities and are often designated as lap burners, While they do employ fans and heat pipes, we are looking at something more effective for the future.

Design

One of the first problems with laptops is that adding something externally to it, or increasing its bulk defeats the purpose of the device. So, it is understandable that manufacturers have focussed heavily on improving the design of the inbuilt fan as a way to target overheating issues. Acer's AeroBlade 3D fan technology changes the typical laptop fan in a number of ways to improve cooling. Using thinner fan blades to incorporate more blades with a curved axial fin on the top of each blade lets it provide 35% more airflow than a typical laptop fan. Another example of a cooling centric design is the Asus ROG Zephyrus that features an 'Active Aerodynamics System' which raises a section of the bottom of the laptop when the lid is opened. This creates a wide, rear open vent through which the fans can push the heat being generated. There are several other examples where the design of a laptop has played an important role in its thermal management.



Aeroblade 3D

External attachments

If you're willing to sacrifice on mobility upto some extent, there are quite a few doors that open up for laptops. Cooling pads are an alternative that have been used for a while now, and have faced considerable debate regarding their effectiveness. Typically, cooling pads feature additional fans and are USB powered to improve airflow across the bottom of your laptop. On the liquid cooling side of things, some manufacturers like Asus have gone all the way and incorporated a detachable liquid cooling setup with their ROG GX lineup of laptops. The setup is found in a bulky attachment at the rear of these devices.

The ASUS ROG GX700

ACROSS DEVICES

There are few cooling technologies that are applicable and effective across devices. Take graphene, for instance. Due to its high conductivity, thin structure and inexpensive nature, it forms an ideal candidate as a thermoelectric cooling medium. Additionally,scientists at Duke University, along with Intel, have come up with an improved vapour chamber design that incorporates 'jumping droplets' that jump off hydrophobic surfaces. The chamber has a super-hydrophobic floor on one side and a sponge-like ceiling on the other. As the heat increases due to the surrounding electronics, tiny water droplets are formed due to condensation, which fall onto the super-hydrophobic floor. There, they join together into bigger droplets, releasing a small amount of energy that is enough to make it jump off the surface. The water is then soaked back up by the sponge-like ceiling and the whole process starts over.

Similar improvements across all kinds of gadgets are going to keep them from heating up – at least until the next major computational power growth happens. Until then, keep it cool!